

ENHANCED BOILING HEAT TRANSFER WITH COPPER OXIDE HIERARCHICAL SURFACES

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ABSTRACT

We experimentally investigated the enhancement in critical heat flux (CHF) during pool boiling on copper oxide (CuO) hierarchical surfaces, where surface roughness at multiple length scales is presented. On the surface with the highest surface roughness factor of 13.3, a CHF of $\approx 250 \text{ W/cm}^2$ and heat transfer coefficient of $\approx 8 \times 10^4 \text{ W/m}^2 \text{ K}$, were achieved. A good agreement between our developed model and experimental results supports the idea that roughness-amplified surface forces are responsible for CHF enhancement on structured surfaces. The work promises enhanced heat removal capability using hierarchically-structured surfaces for high heat flux thermal management.

KEYWORDS

hierarchical surface, critical heat flux, copper oxide, surface roughness, nanostructure, boiling

INTRODUCTION

Thermal management using two-phase cooling has received significant interest for various high flux applications including concentrated photovoltaics, GaN power amplifiers, and integrated circuits [1-4]. In a boiling heat transfer system, the critical heat flux (CHF), where a vapor film will begin to cover the heated surface leading to a loss of contact between the surface and liquid, represents the operational limit due to the significant reduction in heat transfer efficiency. Therefore, methods to extend CHF have been studied extensively owing to its significant practical importance in high performance thermal management systems [5-11]. Our recent work utilized well-defined silicon pillar arrays to demonstrate the role of increasing surface roughness ratio, *i.e.*, $r = \text{total surface area/projected surface area}$, for CHF enhancement [12] on complete wetting surfaces, where the apparent contact angle is $\beta = 0$. The developed CHF model, based on roughness-amplified capillary forces pinning the contact line of vapor bubbles, showed good agreement with experimental data for microstructured surfaces with $1 \leq r \leq 6$, with further CHF enhancement predicted for higher values of r . In this work, we fabricated copper oxide (CuO)-based hierarchical surfaces with $r = 4.8 - 13.3$ to further increase CHF and support the concept that introducing hierarchy produces a multiplicative effect on contact line pinning forces. Accordingly, we demonstrated $q_c'' = 250 \text{ W/cm}^2$ with wall superheat of $\sim 33 \text{ K}$ on the roughest sample tested, representing a $\sim 200\%$ increase in CHF compared to smooth SiO_2 reference surfaces. The obtained CHF values on the hierarchical surfaces showed good agreement with the model prediction, which supports our physical view of

the enhancement phenomenon and the multiplicative effect of roughness at distinct length scales. This predictable high heat removal capability using scalable fabrication techniques promises an exciting opportunity for new surface designs for high flux thermal management.

CHF MODEL

In our recent study, the CHF condition was predicted based on a force balance between the evaporating vapor momentum, buoyancy, and surface forces acting on the liquid/vapor interface of an individual bubble [5]. We modified this model, however to decouple the bubble geometry from the surface force. Accordingly, the surface force maintaining the position of the contact line is amplified due to a longer effective contact line length on the structured surfaces, leading to an expression of the following form [12]:

$$q_c'' = K \times h_{fg} \rho_g^{1/2} [\sigma_{lv} g (\rho_l - \rho_g)]^{1/4}, \quad (1)$$

where

$$K = \left(\frac{1 + \cos \beta}{16} \right) \left[\frac{2(1 + \alpha)}{\pi(1 + \cos \beta)} + \frac{\pi}{4} (1 + \cos \beta) \cos \psi \right]^{1/2},$$

$\alpha = r \cos \theta_{rec}$ is the amplified surface force term, θ_{rec} is the liquid receding angle on the corresponding smooth surface (*i.e.*, intrinsic receding angle), σ_{lv} is the liquid-vapor surface tension, and ψ is the inclination angle of the surface (*i.e.*, $\psi = 0$ for a horizontal upward facing surface). In the superhydrophilic wetting regime ($\beta = 0^\circ$), Equation 1 shows a proportional increase in CHF with the parameter K , which, in turn, is proportional to the square root of the roughness factor r through α , *i.e.*, $q_c'' \propto \sqrt{r}$. Therefore, further enhancement in CHF should increase monotonically with increasing roughness factor. Indeed, we demonstrated that experimental pool boiling data on microstructured surfaces with roughness factors r ranging from 1.8 to 6 follows this scaling, with reasonable quantitative agreement despite several simplifying assumptions used in the model development [12].

FABRICATION AND EXPERMENTS

Fabrication

We fabricated CuO-based, superhydrophilic hierarchical surfaces with two distinct length scales to further demonstrate our understanding of roughness effects on CHF and to achieve larger heat transfer enhancements. The surfaces were fabricated by electroplating Cu micropillars followed by a chemical oxidation step to form CuO nanostructures [13], as shown

in Fig.1. The electroplated square array of Cu micropillars had heights ranging from 35–68 μm , diameters of 30–35 μm , and spacings of 30 μm . Scanning electron micrographs (SEMs) representative of the realized hierarchical surfaces are shown in Fig. 2(a-d). Also, shown in Fig. 2(e) is a cross-section image of the hierarchical structures obtained using focused ion beam (FIB) milling. Finally, on all surfaces, a 1 μm thick layer of Cu was deposited on the back side of the silicon substrates to facilitate solder attachment of the samples to the test setup. Upon dicing, the samples had a projected surface area of $2 \times 2 \text{ cm}^2$, which is large enough to be considered representative of an infinite plate for water [14, 15] and is of comparable size to typical high heat flux electronic components. The heat transfer performance of the hierarchical surfaces was measured using an experimental pool boiling setup, as shown in Fig. 3 [12]. All tests were performed using degassed, high purity water (Chromasolv for HPLC, Sigma-Aldrich) to avoid premature bubble formation and minimize surface contamination.

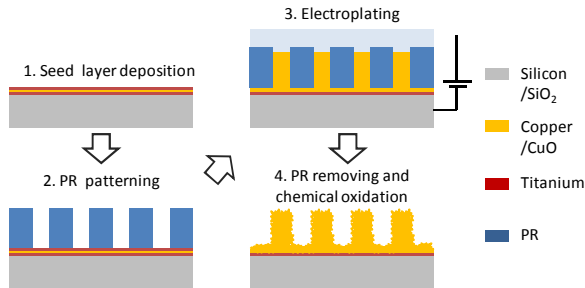


Figure 1: CuO hierarchical surfaces were fabricated as follows: (1) Ti/Cu/Ti layers were deposited on a silicon substrate. (2) A thick ($\approx 100 \mu\text{m}$) PR mold was patterned using photolithography. (3) The top Ti layer was removed to expose the Cu layer and then Cu micropillars were electroplated. (4) After removing the photoresist, Cu micropillars were oxidized in an alkali solution composed of NaClO_2 , NaOH , $\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}$ to form CuO nanostructures.

Surface Characterization

To estimate the surface roughness factor, r , for the CuO surfaces, we first characterized the roughness factors of the nanoscale structure component, r_n . However, the roughness by direct methods such as AFM, FIB images, and contact angle measurement [16], did not reflect how the vapor bubble was in contact with the surfaces, *i.e.*, true contact line length [17]. Therefore, we extracted the r_n of the CuO nanostructures from CHF data obtained on a nanostructured CuO surface using our CHF model in Eq.1. The effective r_n on our CuO nanostructured surfaces estimated by this indirect approach was ≈ 4.8 . We note that the CuO nanostructures on both the smooth and microstructured Cu surfaces were formed using the same oxidation conditions. The total roughness factors, r , were then calculated as the product of r_n and roughness factor of the micropillars r_m (*i.e.*, $r = r_n \times r_m$). The CuO hierarchical surfaces were superhydrophilic behavior at room temperature due to the large roughness factors obtained, $r > 4.8$, and the high surface energy.

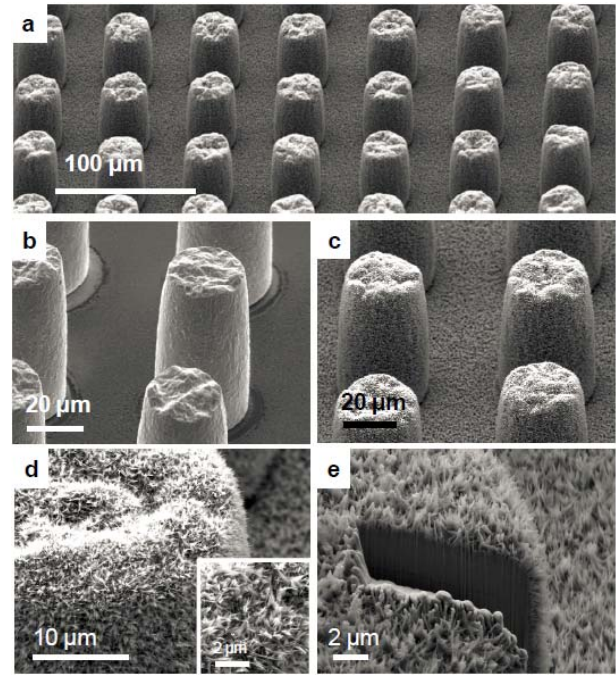


Figure 2: Scanning electron micrographs (SEMs) of the representative, fabricated CuO-based hierarchical surfaces. (a) CuO micropillar array in a square array with heights of 61 μm , diameters of 30 μm , and spacings of 30 μm . (b) Electroplated Cu micropillars, and (c) micropillars with CuO nanostructures formed on the surfaces. (d) Magnified view of the CuO micropillar and CuO nanostructures formed on the surfaces (inset). (e) Cross-sectional view of the CuO micropillar obtained using FIB milling.

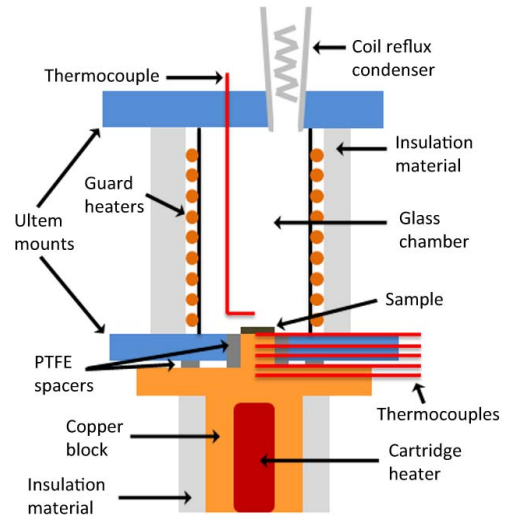


Figure 3: Schematic of the experimental pool boiling setup [12]. The setup consists of a copper block and a glass chamber fixed at both ends by Ultem mounts. Cartridge heaters were imbedded in the copper block allowing for a maximum power of 1400 W. Five in-line thermocouples were inserted into the center axis of the copper block to measure the heat flux and surface temperature.

Experimental Results

Figure 4(a) shows the heat flux q'' as a function of wall superheat $\Delta T = T_w - T_{sat}$, where T_w is the heated

surface temperature and T_{sat} is the saturation temperature (*i.e.*, boiling curve) for the Cu-CuO-based hierarchical surfaces (Hier). Reference boiling curves obtained for smooth SiO₂ surfaces (Sm) and microstructured silicon surfaces (Micro) are also shown in Fig. 4(a) for comparison. Details of the tested surface geometries are listed in Table I. The maximum uncertainty of the heat flux and temperature measurements was calculated to be $\approx 5.6\%$ and ± 1.8 K, respectively [12, 17]. While a CHF of ≈ 83 W/cm² and ≈ 208 W/cm² were obtained on the silicon smooth (Sm) and microstructured (Micro3) surfaces, respectively, CHF values of 250 W/cm² was demonstrated on the best performing Cu-CuO-based (Hier3) hierarchical surfaces. The significant enhancement in CHF on the hierarchical surfaces (up to 200%) is attributed to the high surface roughness factor, which provides a large surface force to balance the momentum force due to evaporation [12]. In addition, the sudden reduction in superheat ΔT along the boiling curve of hierarchical surfaces (*i.e.*, “kickback”) is indicative of nucleation sites within the nanostructures becoming active [18]. This decrease in superheat ΔT but continuous increase in heat flux q'' results in the enhancement in heat transfer coefficient, h near CHF, as shown in Fig. 4(b), representing higher heat transfer performance. Compared to h on smooth (Sm) and microstructured (Micro) surfaces, the CuO hierarchical surfaces demonstrated a significant enhancement in h at high q'' , which was due to the delay in CHF.

TABLE I: Geometric parameters of the hierarchical surfaces. The units of height h , diameter d and (center-to-center) spacing s for micropillars are in microns (μm). The roughness factor of the microstructure, r_m , is calculated with $r_m = 1 + \pi dh / (d + s)^2$. The effective roughness factor of contact line, r , is the products of roughness factors at micro and nanoscales, *i.e.*, $r = r_m \times r_n$.

Sample No.	h	d	s	r_m	r_n	r
Sm	n/a	n/a	n/a	1.0	1.0	1.0
Micro1	20	10	15	2.6	1.0	2.6
Micro2	20	10	10	3.5	1.0	3.5
Micro3	20	10	5	5.4	1.0	5.4
CuO	n/a	n/a	n/a	1.0	4.8*	4.8
Hier1	35	30	30	1.91	4.8*	9.2
Hier2	61	30	30	2.59	4.8*	12.4
Hier3	68	35	30	2.78	4.8*	13.3

*Estimation based on the CHF model and experimental CHF values on CuO nanostructured surfaces

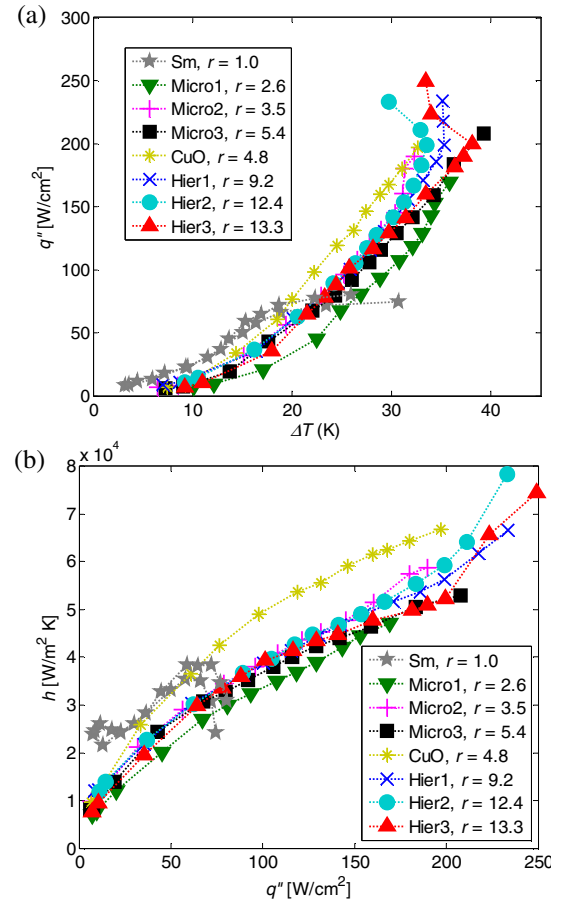


Figure 4. Heat transfer performance. (a) CHF and (b) corresponding heat transfer coefficient comparing smooth (Sm) and microstructured silicon (Micro) and CuO hierarchical (Hier) surfaces

In Fig. 5, the predicted CHF as a function of α (Eq. 1) is overlaid with data from our experiments for the hierarchical surfaces (in red) and previously tested microstructured surfaces [12] (in blue). The dashed line represents the CHF predicted by the classical Kutateladze-Zuber (K-Z) model [19, 20] (hydrodynamic instability mechanism) using an empirical factor of $K = 0.18$ in Eq. 1 [21] obtained from classical pool boiling experiments on unstructured, well-wetting surfaces ($\alpha \approx 1$). The good agreement between the data and model, which does not contain any fitting parameters, for α ranging from 1 to ≈ 13.3 , demonstrates the validity of the model for the structured surfaces in the complete wetting regime. The trend of increasing CHF with increasing surface roughness is well-captured by the model which suggests that the key physics of the CHF mechanism on these structured surfaces were accounted for. While the value of r_n of the CuO surfaces was approximated based on the CHF model and experimental data of CuO nanostructured surfaces, the agreement between the CHF on CuO hierarchical surfaces (Hier1-3) and the model prediction was consistent, suggesting that the effective surface roughness pinning the vapor bubble based on the indirect approach for our CuO nanostructured surfaces provided a reasonable estimation.

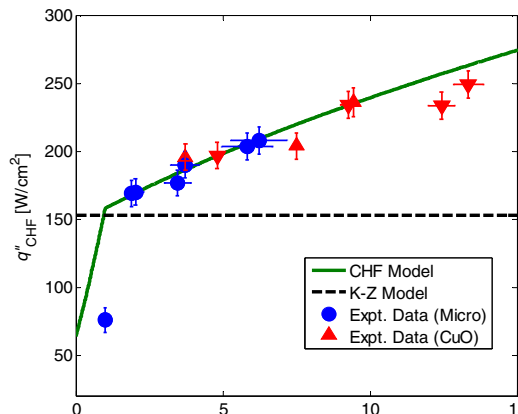


Figure 5. CHF as function of α ($= r \cos \theta_{rec}$). The proposed model (solid line) is compared to the K-Z model with a factor of $K = 0.18$ (dashed line). The symbols show the CHF data for (●) microstructured surfaces [12], and (▼) CuO hierarchical surfaces, as a function of the estimated α values.

CONCLUSION

In conclusion, we fabricated hierarchically-structured surfaces using electroplated and oxidized copper with a roughness factor, r , of $\approx 4.8 - 13.3$ to investigate CHF on high roughness factor surfaces in pool boiling. The good agreement between the CHF model and experimental observations on the hierarchical surfaces indicates that the roughness-amplified surface force plays the defining role in CHF enhancement on structured surfaces with a roughness factor r ranging from 1 – 13.3 with distinct roughness length scales introducing a multiplicative effect on the amplified surface force. A CHF of 250 W/cm^2 (i.e., a $\approx 200\%$ CHF enhancement compared to smooth SiO_2 surfaces) achieved on CuO-based hierarchical surfaces demonstrates high heat removal capability. Furthermore, the scalable fabrication process of electroplating copper micropillars coupled with a simple chemical oxidation process promises an exciting opportunity to achieve high performance boiling heat transfer.

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